

Xist RNA binds select autosomal genes and depends on Repeat B to regulate their expression

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Abstract

Xist, a pivotal player in X chromosome inactivation (XCI), has long been perceived as a cis-acting long noncoding RNA that binds exclusively to the inactive X chromosome (Xi). However, Xist's ability to diffuse under select circumstances has also been documented, leading us to suspect that Xist RNA may have targets and functions beyond the Xi. Here, using female mouse embryonic stem cells (ES) and mouse embryonic fibroblasts (MEF) as models, we demonstrate that Xist RNA indeed can localize beyond the Xi. However, its binding is limited to ~100 genes in cells undergoing XCI (ES cells) and in post-XCI cells (MEFs). The target genes are diverse in function but are unified by their active chromatin status. Xist binds discretely to promoters of target genes in neighborhoods relatively depleted for Polycomb marks, contrasting with the broad, Polycomb-enriched domains reported for human XIST RNA. We find that Xist binding is associated with down-modulation of autosomal gene expression. However, unlike on the Xi, Xist binding does not lead to full silencing and also does not spread beyond the target gene. Over-expressing Xist in transgenic ES cells similarly lead to autosomal gene suppression, while deleting Xist's Repeat B motif reduces autosomal binding and perturbs autosomal down-regulation. Furthermore, treating female ES cells with the Xist inhibitor, X1, leads to loss of autosomal suppression. Altogether, our findings reveal Xist targets ~100 genes beyond the Xi, identify Repeat B as a crucial domain for its in-trans function in mice, and indicate that autosomal targeting can be disrupted by the X1 small molecule inhibitor.

eLife Assessment

This **valuable** study addresses potential roles of the master regulator of X chromosome inactivation, the Xist long non-coding RNA, in autosomal gene regulation. Using data from mouse cells, the authors propose that Xist can coat specific autosomal promoters, which in turn leads to the attenuation of their transcriptional activity, complementing recently published results from humans. While the evidence from individual genes is suggestive, shortcomings in the data and statistical analyses leave the evidence currently **incomplete**. The work would be of interest to anyone studying gene regulation and noncoding RNA.

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Introduction

The unequal sex chromosome composition between male (XY) and female (XX) placental mammals necessitates dosage compensation of the X chromosome ^{1,2}. X chromosome inactivation (XCI) specifically evolved to ensure equal X-linked gene dosage between the sexes. During early embryogenesis, one of the two X chromosomes in female cells is randomly silenced, leading to a mosaic of cells in which X-linked genes can be expressed from either the maternal or paternal X chromosome ³⁻⁵. The long non-coding RNA Xist is an essential regulator of the XCI process ^{6,7}. At the onset of XCI, Xist is expressed exclusively from the future inactive X chromosome (Xi) and then selectively spreads *in cis* along the chromosome to initiate formation of heterochromatin via association with chromatin-modifying complexes and alterations in 3D chromosome structure (reviewed in ^{2,8,9-12}). From early RNA fluorescence in situ hybridization (FISH) experiments, it was shown at a cytological level that Xist binds only to the X-chromosome which transcribes the RNA ^{6,13}. In Xist transgenesis studies, the RNA is also observed to localize exclusively *in cis* to the transgene, even on autosomes ¹⁴⁻²⁰. In more recent years, technical advances in epigenomic mapping of RNA confirmed the early cytological data and provided a map of Xist binding sites *in cis* at kilobase resolution ^{21,22}. Furthermore, genetic analysis of the X-inactivation center revealed a Xi-specific nucleation site for the initial binding of Xist that then enabled the RNA to spread exclusively *in cis* ²³. Altogether, these studies solidified the view that Xist RNA is a *cis*-acting RNA.

However, studies have long documented the potential for Xist to spread beyond the X-chromosome under various non-physiological conditions. When Xist is overexpressed, the RNA can diffuse to bind neighboring chromosomes ^{15,23,24}. Furthermore, when Xist's nucleation site is mutated, Xist will diffuse and bind other chromosomes *in trans* ²³. A recent study has also pointed to broad XIST binding patterns outside of the Xi in human naïve stem cells and suggested autosomal targets ²⁵. These findings prompt interesting questions about the binding dynamics and properties of Xist on autosomes, its potential to spread locally in autosomal domains, and its impact on gene regulation and cellular physiology. To investigate, here we examine the capacity for Xist to spread beyond traditional boundaries. We use mouse embryonic stem cells (mESC) and mouse embryonic fibroblasts (MEF) in order to capture both the establishment and maintenance phases of XCI. Intriguingly, we identify about 100 binding sites on autosomes in cells undergoing XCI as well as post-XCI cells. We demonstrate discrete binding sites, rather than broad binding domains. Transcriptomic analysis reveals a selective downregulation, but not silencing, of associated genes. We also probe a requirement for Xist's Repeat B (RepB) motif and the ability to perturb autosomal effects by treating cells with a small molecule inhibitor of Xist RNA.

Results

Epigenomic mapping of Xist binding reveals autosomal targets in mouse cells

To explore the possibility of Xist extending its binding to autosomal sites beyond the X chromosome, we conducted CHART (Capture Hybridization of Associated RNA Targets) ^{26,21} in female ES cells. To capture Xist RNA, we used oligoprobes antisense to Xist and then pulled down interacting chromatin for deep sequencing of the associated DNA. We use sense probes (complementary to the antisense Xist probes) to control for any direct hybridization to DNA that could occur independently of Xist RNA. This helps us distinguish signals genuinely due to Xist RNA from those that could arise from non-specific probe binding to DNA. Also, we performed CHART using male ES cells, which lack Xist expression, to define the background level of binding. These

two controls allowed us to account for non-specific interactions and background noise. In addition to these controls, we performed 2 biological replicates and used the overlapping regions of these 2 replicates to ensure the reliability and reproducibility of our results. We normalized our data to input DNA to account for differences in chromatin preparation and sequencing depth.

Xist RNA is transcriptionally upregulated during female ES cell differentiation when XCI is induced. To profile the dynamics during cell differentiation and XCI, we examined wild-type (WT) female ES cells in the undifferentiated state (day 0) and at three differentiated stages (day 4, 7, and 14). Antisense oligonucleotides targeting Xist efficiently pulled down Xist RNA and associated chromatin targets, in excess of the “background” observed with the sense probe, and male control (**Figure S1A**). As expected, the Xist CHART signal in WT female ES cells displayed strong binding at the Xist locus and genes subject to XCI, such as *Cdkl5* and *Mecp2*, while showing significantly weaker binding at the escapee gene, *Kdm6a*. In contrast, CHART experiments using sense probes and male ES cells, which lack Xist expression, showed minimal binding to these genes (**Figure S1B-S1E**). Because Xist spreads *in cis* along the Xi to cover much of the chromosome²⁷, we first examined total Xist coverage (normalized to input), rather than calling peaks. Day 0 ES cells showed no enrichment of Xist binding, consistent with Xist being in the uninduced state (**Figure 1A**). Upon cell differentiation, enriched binding of Xist was consistently observed across all time points (**Figure 1A**). Greatest enrichment on the X chromosome was observed on day 7, in agreement with *de novo* XCI when Xist is needed at highest concentration^{21,28}.

Intriguingly, on day 7, only ~10% of the reads mapped to the X chromosome (**Figure S1A**), prompting the question of whether Xist might have non-X-linked targets as well. Indeed, although the X chromosome was most enriched for Xist, we noticed that autosomes also showed highly reproducible peaks, beginning at day 4 and increasing across the differentiation time points (**Figure 1A**). In contrast to broad binding pattern covering the entire X chromosome (**Figure 1B**), Xist binding patterns on autosomes trended towards sharp peaks (**Figure 1C** and **1D**). To determine if the peaks were statistically significant, we called Xist peaks using MACS2 on the Xist CHART data. We restricted the analysis to autosomal reads to increase the sensitivity of the analysis. Intriguingly, we found hundreds of significant peaks across autosomes between days 4-14 of differentiation (**Figure 1E-1G**, **Figure S2A**, Tables S1-S6). Notably, Xist binding sites on autosomes were disproportionately located in promoter regions (**Figure 1A** and **1H**), hinting at a potential role for Xist in autosomal gene regulation. We conclude that, in addition to the Xi of differentiating ES cells, mouse Xist RNA selectively binds ~100 autosomal targets, preferentially at promoter regions. This promoter-dominant profile contrasts with Xist patterns identified in human cells, which tend to demonstrate broad regions of coverage over genes²⁵. Although the Gene Ontology (GO) analysis of these Xist autosomal target genes did not yield statistically significant enrichment for specific biological processes or pathways, a closer examination reveals that these genes have diverse functions. They are involved in processes such as cancer development (*Bcl7b*), POLII transcription regulation (*Med16*), amino acid transport (*Slc36a4*), RNA binding (*Rbm14* and *Stau2*), among other processes. Together with previously published work in human cells²⁵, our findings suggest that Xist may have a broad regulatory impact on a variety of autosomal genes, extending its influence beyond X-chromosome inactivation.

Deleting RepB causes a loss of binding to autosomal target genes

Prior work showed that Xist’s Repeat B (RepB) element is essential for Polycomb recruitment²⁹⁻³². RepB is also required for proper spreading and localization of Xist RNA to the Xi: Without RepB, the Xist RNA cloud was observed to adopt a dispersed appearance consistent with diffusion of the RNA away from the Xi³¹. Given the partial loss of attachment to the Xi, here we asked if the Xist diffusion could lead to enhanced binding to autosomal targets. We conducted CHART experiments in RepB deletion (Δ RepB) female ES cells at various differentiation stages and quantitated the degree of X-linked versus autosomal binding (**Figure 2**). Although Xist retained partial binding to the X-chromosome, there was a significant decrease in localization along X-linked genes along all differentiation days, especially at day 7 (**Figure 2A**), consistent with RepB

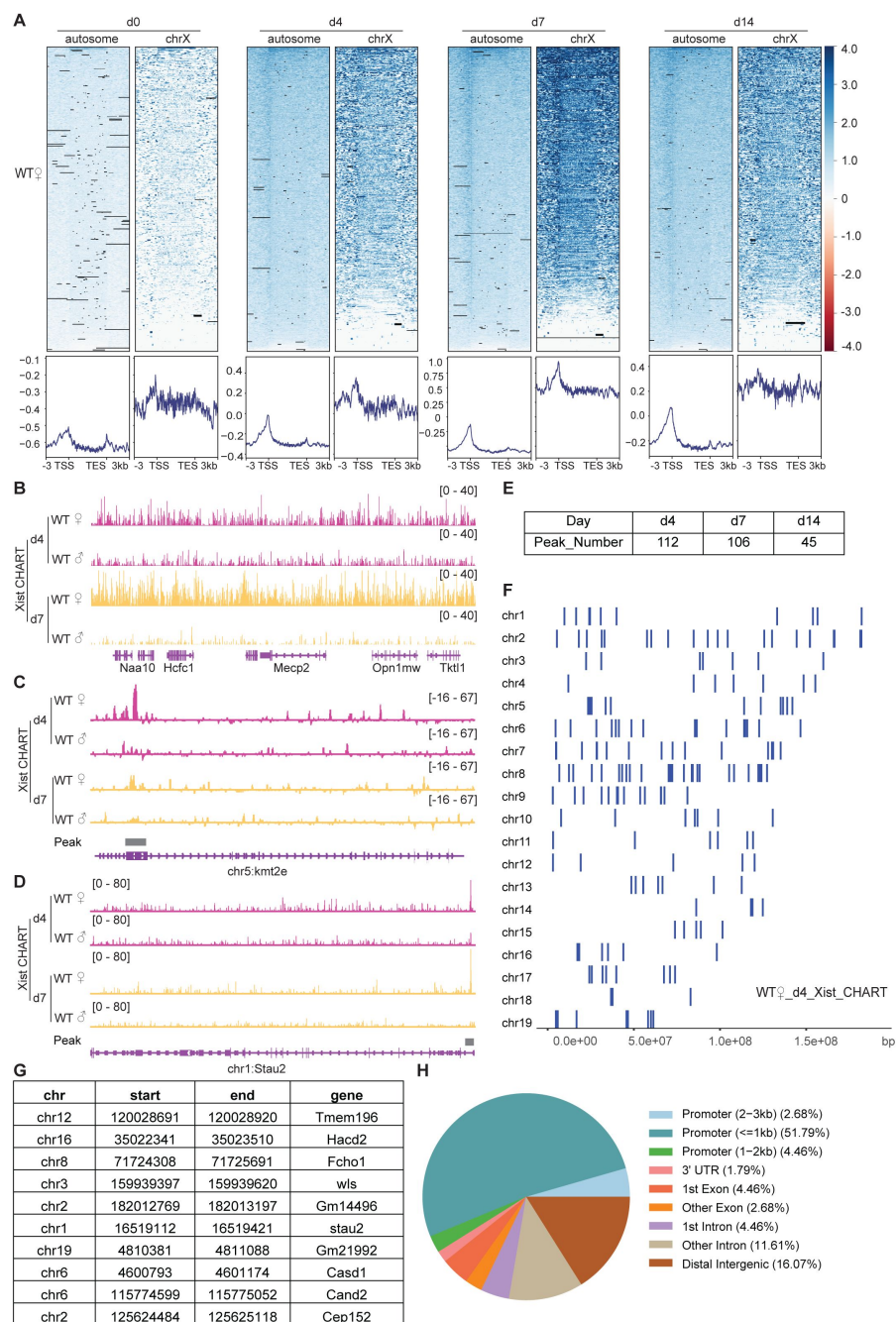


Figure 1.

CHART-Seq reveals ~100 discrete binding sites in autosomal genes.

- A.** Heatmap and profile of Xist coverage on autosomes and chromosome X genes in WT female ES cells at day 0, day 4, day 7, and day 14.
- B.** Representative consecutive binding signals of Xist on chromosome X (~320 kb) in WT female ES cells on day 4 and day 7. WT male ES cells are used as control.
- C-D.** Two representative site-specific binding of Xist on autosome locus in WT female ES cells on day 4 and day 7, WT male ES cells are used as control.
- E.** Number of Xist peaks determined by MACS2 peak calling on autosomes in WT female ES cells at day 4, day 7, and day 14.
- F.** Distribution pattern of Xist peaks identified by MACS2 peak calling on autosomes in WT female ES cells.
- G.** 10 examples of Xist binding position information and related genes on autosomal chromosomes in WT female ES cells.
- H.** Feature annotation of Xist binding loci on autosomes in WT female ES cells.

being essential for attachment of Xist to the Xi³¹. Intriguingly, however, the loss of Xi binding was not accompanied by any significant increase in Xist coverage at the same autosomal targets, at any differentiation day (**Figure 2A**). In fact, there appeared to be decreased binding to the ~100 autosomal genes. However, there was not a change up-or down-wards in the number of autosomal targets (**Figure 2B-2D**, S2A, Tables S1-S6). A similar number of Xist peaks across autosomes in Δ RepB cells was observed and the autosomal targets remained similar. Moreover, in the absence of RepB, Xist retained a preference to bind promoter regions (**Figure 2E-2H**). Thus, RepB is required not only for Xist to localize to the X-chromosome but also for its localization to the ~100 autosomal genes.

Xist RNA down-regulates autosomal genes in a RepB-dependent manner

The promoter-specific binding pattern of Xist on autosomal targets contrasts sharply with the broad binding pattern of Xist on the Xi (**Figure 1A**) and with broad regions identified for human autosomal genes²⁵. Given the preference for autosomal promoters, we investigated whether Xist modulates expression of the associated autosomal genes. To address this question, we conducted transcriptome sequencing on WT female ES cells across day 0, 4, 7, and 14, and compared the profile to those of Δ RepB female ES cells. To analyze the characteristics of genes bound by Xist on autosomes, we categorized all refSeq genes based on their expression levels at each differentiation day. The genes were divided into 5 quintiles (Q1 silent, Q2 low, Q3 moderate, Q4 high, Q5 highest). Consistent with Xist's behavior on the X-chromosome^{21,22}, Xist also favored binding to actively expressed genes on autosomes (**Figure 3A**). To examine gene expression levels as a function of distance from the Xist binding site, we divided neighboring genes into 10-to 100-kb bins and observed that genes at <10 kb showed highest Xist levels (**Figure 3B**, **Figure S2B and S2C**). These findings support Xist's favoring actively expressed genes. We then examined Polycomb marks in relation to the autosomal Xist domains. In agreement with the Xist's predilection for active genes, we observed lower coverage of Polycomb marks associated with Xist RNA^{12,17,29-33}, H3K27me3 and H2AK119ub (associated with PRC2 and PRC1 respectively), as shown by lower coverages within the 10-kb versus 50-kb neighborhoods (**Figure 3C**, **Figure S2D**). These findings also contrasted with observations in human cells, where PRC2/H3K27me3-enriched regions were favored²⁵. Our findings indicate that WT Xist transcripts favor binding to a select set of actively transcribed autosomal genes and these genes are not marked by Polycomb in mouse cells.

Given the association with active autosomal genes, we next asked whether Xist's binding modulates their expression. We performed RNA-seq analysis in two biological replicates in the ES differentiation time series. First, we profiled gene expression in female ES cells relative to male ES cells, reasoning that the lack of Xist RNA in male ES cells might lead to a difference in female versus male expression of autosomal target genes. Indeed, among the ~100 target genes, there was significantly greater expression in male cells (**Figure 3D-3H**), suggesting that Xist might also negatively regulate the select genes on autosomes. To test this idea, we took advantage of the Δ RepB mutation, as RepB is required for proper silencing of Xi genes³¹. Transcriptomic analysis demonstrated that perturbing RepB resulted in loss of repression of autosomal targets. A detailed examination focusing on the top 100 peaks of Xist-bound genes on autosomes unveiled a down-modulation especially evident on days 4 and 7 of ES cell differentiation (**Figure 3G** and **3H**, **Figure S3A-S3D**). The Xist-bound autosomal genes on day 14 of ES cell differentiation like *Ces1l* or non-targets of Xist on autosomes such as *Kmt2c* did not demonstrate significant changes in gene expression (**Figure S3E and S3F**). We note, however, that unlike on the Xi, autosomal binding of Xist did not lead to full silencing of the target gene. Rather, the effect was a partial repression. The loss of repression tracked with rescued Xist binding to the autosomal genes (**Figure 3F** and S3A-S3D). To confirm that the observed changes in autosomal genes are due to Xist binding, we performed a similar analysis on 100 randomly selected autosomal genes that did not have Xist

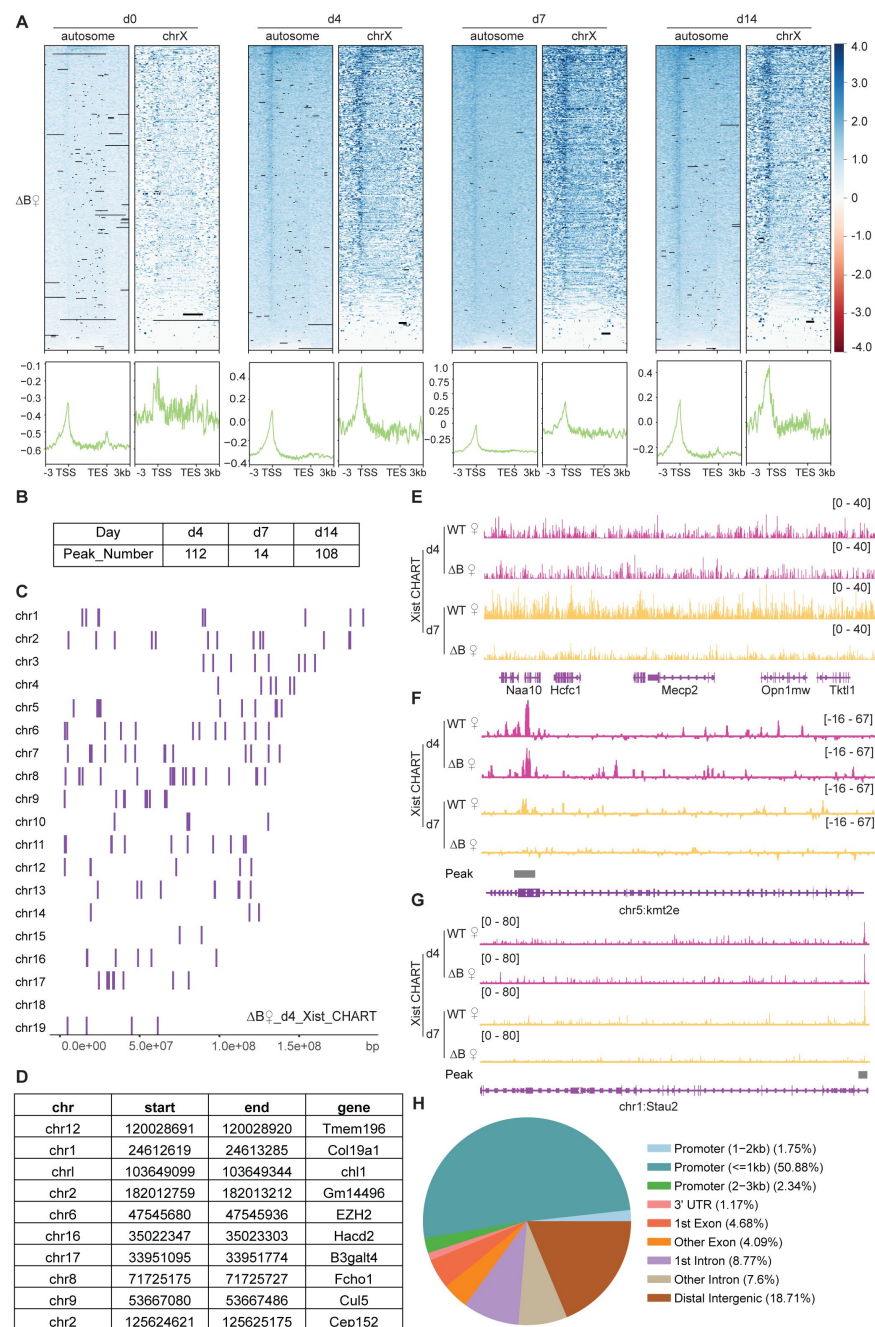


Figure 2.

Xist's Repeat B is required for proper binding of Xist to autosomal genes.

A. Heatmap and profile of Xist coverage on autosomes and chromosome X genes in Δ RepB female ES cells at day 0, day 4, day 7, and day 14.

B. Number of Xist peaks determined by MACS2 peak calling on autosomes in Δ RepB female ES cells at day 4, day 7, and day 14.

C. Distribution pattern of Xist peaks identified by MACS2 peak calling on autosomes in Δ RepB female ES cells.

D. 10 examples of Xist binding position information and related genes on autosomal chromosomes in Δ RepB female ES cells.

E. Representative consecutive binding signals of Xist on chromosome X (~320 kb) in WT and Δ RepB female ES cells on day 4 and day 7.

F-G. Two representative site-specific binding of Xist on autosome locus in WT and Δ RepB female ES cells on day 4 and day 7.

H. Feature annotation of Xist binding loci on autosomes in Δ RepB female ES cells.

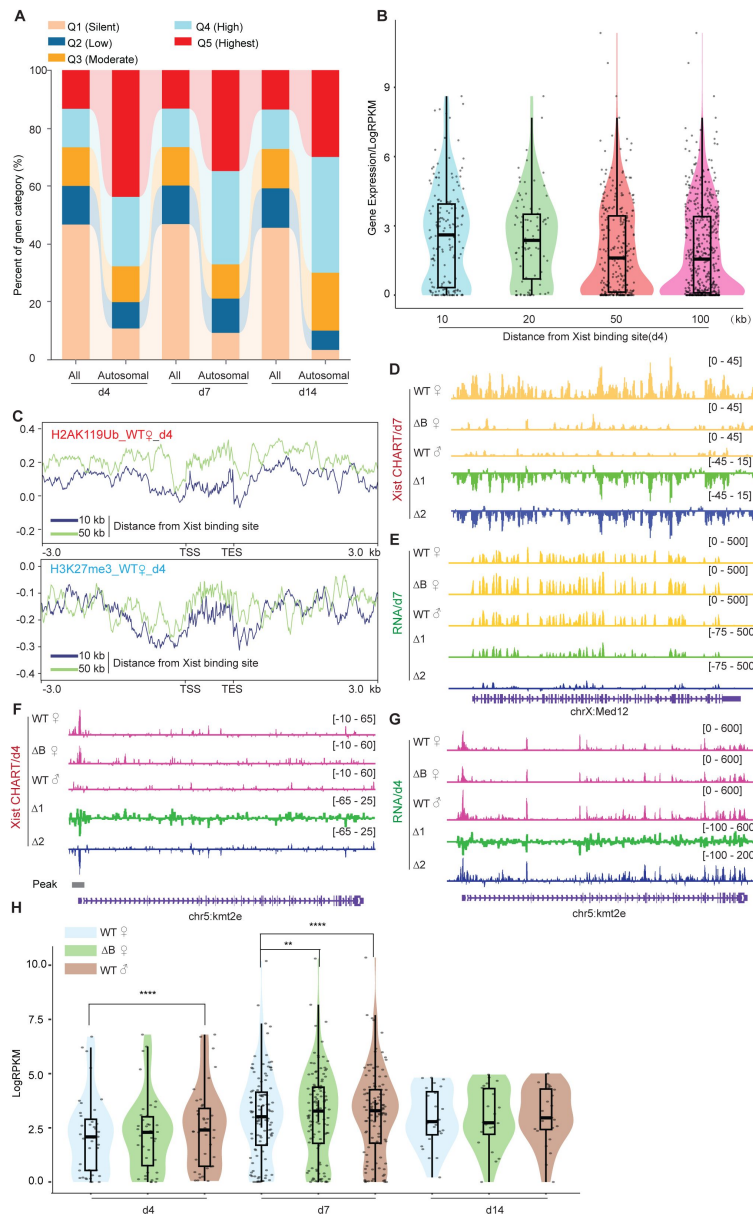


Figure 3.

Autosomal target genes are actively transcribed.

A. Xist preferentially targets actively expressed genes. The proportion of overall genes (label as “All”) and Xist targets on autosomes expression levels (label as “Autosomal”) in different expression categories. Genes are classified into distinct categories (Q1-Q5) based on RPKM values, representing various levels of expression. Category Q1 includes non-expressed genes (RPKM=0), while categories Q2, Q3, Q4, and Q5 represent 25%, 50%, 75%, and 100% expression, respectively.

B. Genes bound by Xist exhibit elevated expression levels compared to surrounding regions. The analysis of gene expression within the 10, 20, 50, and 100-kilobase binding regions of Xist is conducted in WT female ES cells at day 4.

C. The H2AK119ub and H3K27me3 (ChIP-Seq) profile of genes within the 10 and 50-kilobase binding regions of Xist in WT female cells at day 4.

D-E. Exemplifying CHART-seq (D) and RNA-Seq (E) patterns of an X-linked gene (*Med12*) at day 7. Change in coverage ($\Delta 1$ and $\Delta 2$) is shown below ($\Delta 1$ for Δ RepB $\varnothing$ -WT $\varnothing$, and $\Delta 2$ for WT σ -WT $\varnothing$).

F-G. Representing CHART-seq (F) and RNA-Seq (G) patterns of an autosomal gene bound by Xist (*Kmt2e*) at day 4. Change in coverage ($\Delta 1$ and $\Delta 2$) is shown below ($\Delta 1$ for Δ RepB $\varnothing$ -WT $\varnothing$, and $\Delta 2$ for WT σ -WT $\varnothing$).

H. Assessing gene expression levels of Xist targets on autosomes (10 kb within the peak region) in WT, Δ RepB female ES cells, and male ES cells at different time points. P-values are determined using the Wilcoxon rank sum test.

binding (“non-targets”). The non-targets did not show preference for specific autosomal gene region (**Figure S4A-C**) and showed no significant difference in Polycomb enrichment (**Figure S4D**) or gene expression (**Figure S4E**). These results demonstrate that autosomal binding of Xist RNA directly leads to a suppression of gene expression in a RepB-dependent manner.

Effects of Xist overexpression and X1 drug treatment on autosomal genes

To seek further evidence for autosomal effects of Xist binding, we conducted two orthogonal lines of experimentation. First, we asked the reciprocal question and determined whether ectopically overexpressing Xist leads to increased autosomal repression. We examined two transgenic Xist cell lines, Tg_1 and Tg_2³⁴ (**Figure 4A**). These two transgenic (Tg) females cell lines respond to doxycycline to induce Xist expression on autosomes (specifically chromosome 12, with two different transgene insertion sites). Transcriptomic analysis of these two Tg mouse ES cells upon neuronal differentiation showed a significant suppression of X-linked genes (e.g., *Med14*) (**Figure 4B**). Autosomal Xist targets, as exemplified by *Bcl7b* and *Rbm14*, were also significantly suppressed beyond what was observed in WT ES cells (**Figure 4C** and **4D**). In contrast, non-targets of Xist such as *Stau1* did not demonstrate significant changes in gene expression (**Figure 4E** and **4G**). Looking across all autosomal target genes, we observed a significant decrease in mean expression in the Xist overexpressing cell lines (**Figure 4F**), further implicating Xist binding in direct regulation of autosomal gene targets.

In the second line of investigation, we deployed X1, a small molecule inhibitor of Xist that was previously shown to block initiation of XCI in female ES cells through Xist’s Repeat A (RepA) motif³⁵. We reasoned that if the mechanism of autosomal gene suppression were similar to XCI, treating cells with X1 could also blunt autosomal gene suppression. Indeed, just as X-linked genes such as *Mecp2* failed to undergo silencing after 5 days of differentiation, transcriptomic analysis showed that autosomal gene targets also failed to be suppressed in the presence of X1 (**Figure 5A-C**). On average, there was a significant upregulation of autosomal targets on day 5 of differentiation and X1 treatment (**Figure 5E**). Again, non-targets of Xist such as *Cbx2* did not demonstrate significant changes in gene expression (**Figure 5D** and **5F**). Analysis of ChIP-seq data for H3K27me3 indicated a reduction of the PRC2 mark within the Xist target gene and flanking regions (**Figure 5G** and **5H**), consistent with X1 blocking RepA’s recruitment of PRC2³⁵. In contrast, Xist non-target genes show no significant changes in PRC2 signal (**Figure 5I**). These additional perturbation studies reinforce the notion that autosomal binding of Xist RNA directly suppresses autosomal targets. Thus, it is possible to disrupt autosomal Xist binding by administering a small molecule inhibitor of Xist.

Xist RNA also binds to autosomal genes in post-XCI cells

Lastly, because the establishment and maintenance phases of XCI show differential sensitivity to loss of Xist^{36–39}, we asked if autosomal repression exhibits a similar sensitivity. To address this, we turned to mouse embryonic fibroblasts (MEF), where XCI is well established. We first conducted CHART assays to determine whether Xist binds to the Xi and autosomes in a similar way in MEFs (**Figure 6A**). Because the autosomal signals displayed peak-like characters, we used MACS software to call significant peaks on autosomes to determine if there was an increase in promoter-specific binding of the autosomal genes. We observed that, while Xist continued to favor targeting of promoter regions in both WT and ΔRepB MEFs (**Figure 6B-6D**), Xist peaks did not increase in size in ΔRepB MEFs; rather, the peaks decreased in size (**Figure 6B-6D**) — a result similar to that in ΔRepB ES cells (**Figure 2**). Furthermore, there was a considerable decrease in the number of significant peaks in ΔRepB MEFs when compared to WT MEFs (**Figure 6B**). In WT MEFs, Xist clearly favored Xi binding, but Xist signals could also be visualized on autosomes (Table S7-S10, **Figure 6F**), in agreement with the results in ES cells (**Figure 1**). In ΔRepB MEFs, there was a partial loss of Xi localization, aligning with previous findings³¹ (**Figure 6E-6G**).

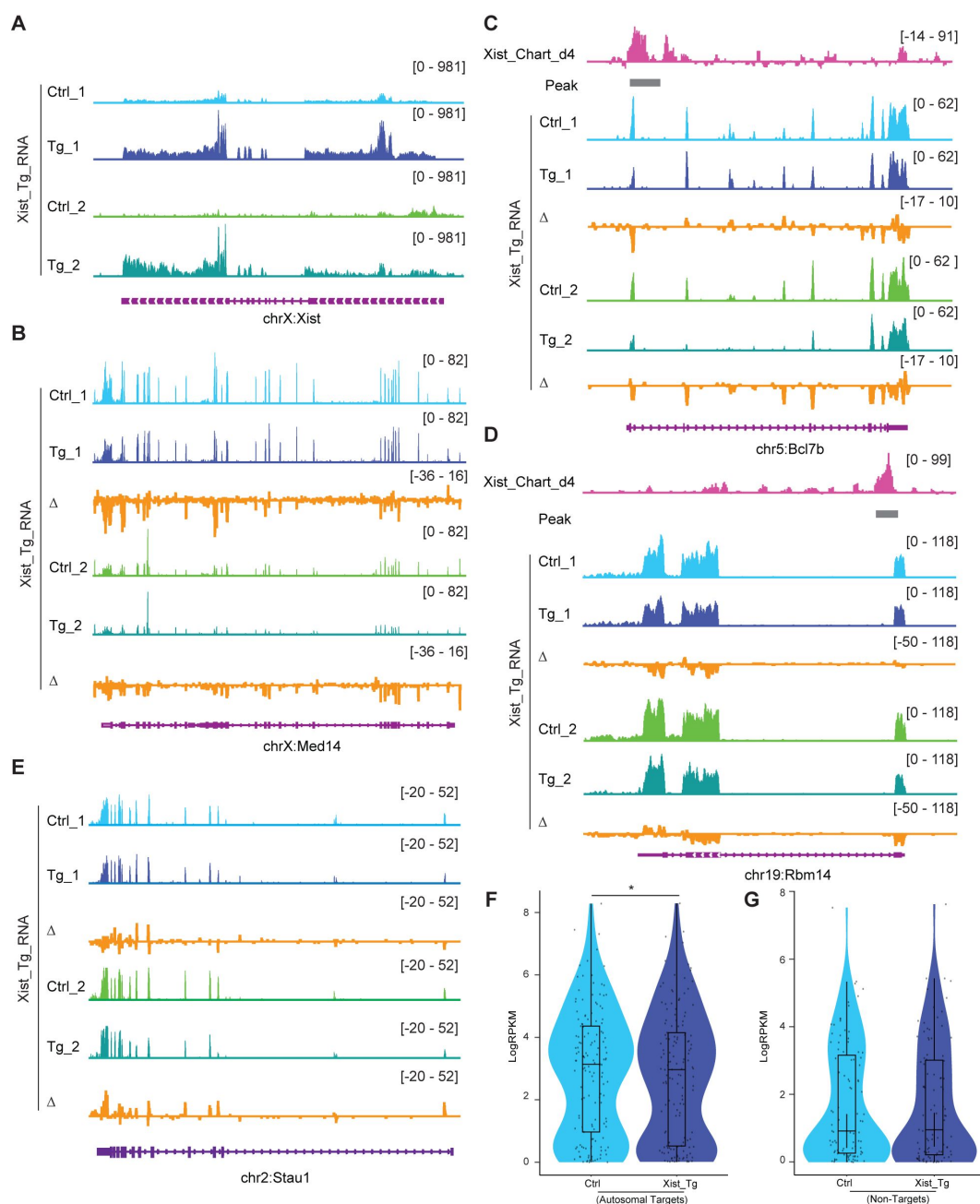


Figure 4.

Xist binding suppresses, but does not silence, autosomal gene expression.

- A.** RNA-seq track shows the Xist expression level in ectopic Xist overexpressed ES cell lines (Tg).
- B.** RNA-seq track shows the MED14 (X-linked gene) expression level in ectopic Xist overexpressed ES cell lines (Tg). Change in coverage (Δ) is shown below (Tg - Ctrl).
- C-D.** RNA-seq track illustrating expression alongside representative autosome genes with Xist binding (*Bcl7B* and *Rbm14*) in differentiated Xist Tg ES cells (neurons). Change in coverage (Δ) is shown below (Tg - Ctrl).
- E.** RNA-seq track illustrating expression level of autosome genes without Xist binding (*Stau1*) in differentiated Xist Tg ES cells (neurons). Change in coverage (Δ) is shown below (Tg - Ctrl).
- F-G.** Analysis of gene expression levels for Xist targets (F) or non-targets (G) on autosomes in differentiated Xist Tg ES cells (neurons). P-values are determined using the Wilcoxon rank sum test.

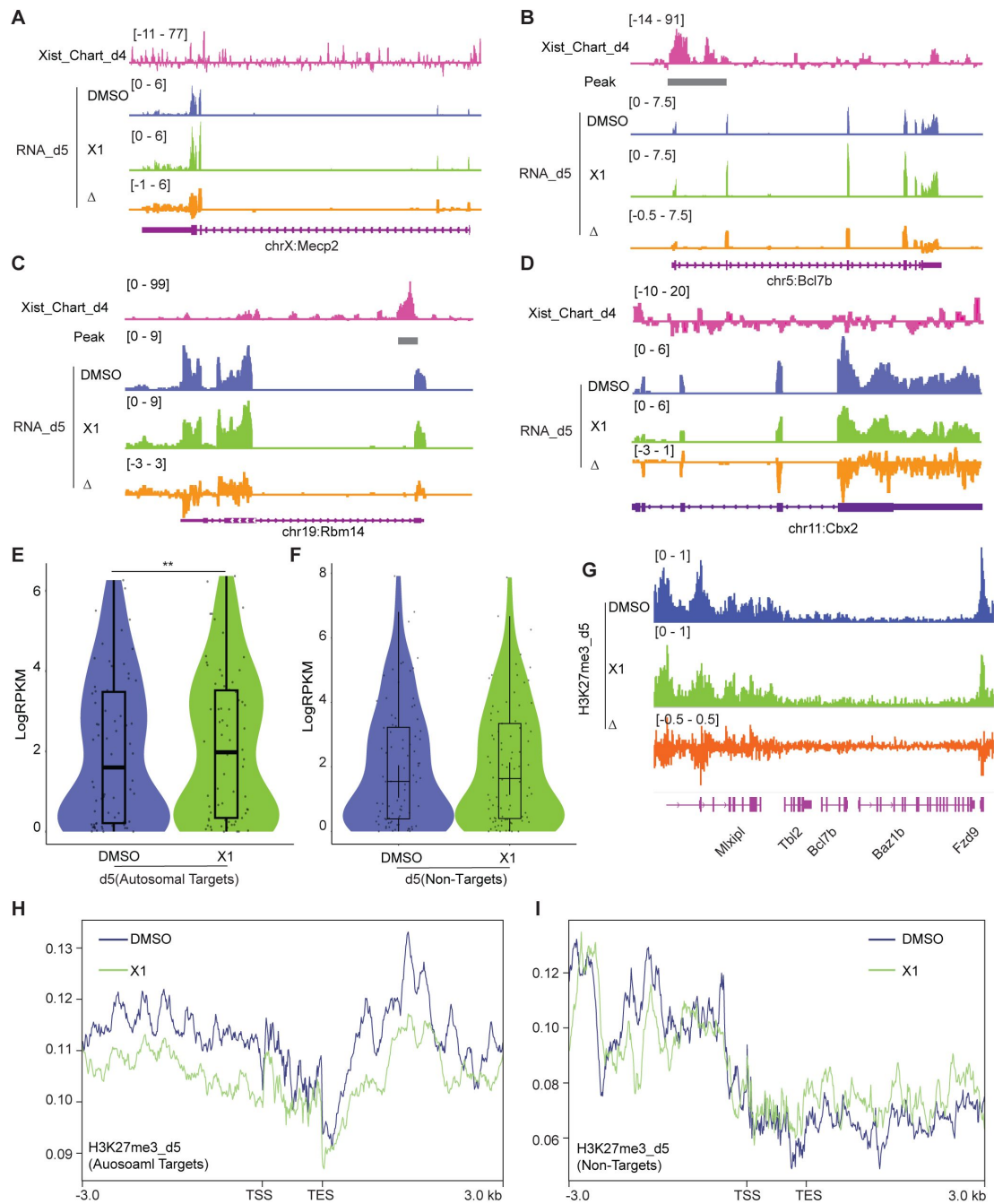


Figure 5.

Treating cells with the X1 inhibitor of Xist RNA perturbs autosomal target genes.

A-D. RNA-seq track depicting *Mecp2* (A) (an X-linked gene), *Bcl7b* (B) and *Rbm14* (C) (Xist-regulated autosome target), and *Cbx2* (D) (autosomal gene without Xist binding) expression in DMSO (control) and Xist inhibitor (X1) treated differentiated WT ES female cells at day 5 of differentiation. Change in coverage (Δ) is shown below (X1 - DMSO).

E. Evaluation of gene expression levels for Xist targets on autosomes in WT and X1 treated ES cells. P-values are determined using the Wilcoxon rank sum test.

F. Evaluation of gene expression levels for Xist non-targets on autosomes in WT and X1 treated ES cells.

G. ChIP-seq track for H3K27me3 on *Bcl7b* (within a 160 kb window) in WT and X1 treated ES cells. Change in coverage (Δ) is shown below (X1 - DMSO).

H-I. Profile plot displaying H3K27me3 levels for Xist-flanking genes (H) (within a 100 kb window) or Xist non-targets (I) in WT and X1 treated ES cells.

Consistent with these findings, RNA-seq analysis of Δ RepB MEFs did not reveal any significant change in expression of Xi genes (e.g., *Mecp2*) and autosomal target genes (e.g., *Rbm14*) (**Figure 7A-7D**).

We then asked if the retention of some Xist binding on the Xi could explain the lack of transcriptomic difference in Δ RepB MEFs. To investigate, we utilized a deletion of Xist's Repeat E (Δ RepE), which was previously demonstrated to severely abrogate localization of Xist to the Xi^{40,41}. We reasoned that the severe loss of Xist binding might unmask a transcriptomic difference. We observed that Xist signals were more substantially reduced on both the Xi and autosomal regions in Δ RepE MEFs compared to Δ RepB cells (**Figure 6E-6G**). Despite this reduction, peak coverages in autosomal target genes did not increase in Δ RepE MEFs (**Figure 6B-6D**). In fact, there was an overall decrease in the number of significant Xist peaks in Δ RepE MEFs as well (**Figure 6B**). Furthermore, we observed no transcriptomic difference in Δ RepE MEFs relative to WT MEFs (**Figure 7A-7E**). Additionally, further examination of RNA sequencing data from male and female MEF cells in two published studies^{42,43} corroborated that the expression levels of these autosomal Xist targets did not exhibit significant changes (**Figure 7F** and **7G**). Altogether, the analysis in MEFs demonstrates that Xist continues to bind autosomal genes in post-XCI somatic cells. However, autosomal binding of Xist in post-XCI cells does not seem to impact expression of the associated autosomal gene.

Discussion

Together with a backdrop of studies on ability of Xist RNA to diffuse and bind chromatin in trans^{15,23-25}, the results of our current work challenges the conventional narrative that Xist operates exclusively *in cis* on the Xi. We find that autosomal Xist targets are not numerous, possibly limited to only ~100 in both pluripotent stem cells and in somatic cells. On autosomes, Xist does not spread and instead covers only a narrow region corresponding to single genes. The genes tend to be active genes, with Xist specifically targeting the promoters of those genes. Genetic analysis coupled to transcriptomic analysis showed that Xist down-regulates the target autosomal genes without silencing them. This effect leads to clear sex difference — where female cells express the ~100 or so autosomal genes at a lower level than male cells (**Figure 7H**). Thus, our findings redefine the scope of Xist's functional repertoire and provide insights into the broader landscape of epigenetic regulation during cellular differentiation. Xist RNA therefore plays a more complex role than previously envisaged, with several implications and caveats.

First, the observed consistent binding of Xist to both the X chromosome and autosomes in female ES cells prompted further exploration of the intricate dynamics of Xist during cellular differentiation. Our categorization of genes based on their expression levels revealed a compelling correlation between Xist binding on autosomes and the expression levels of associated genes. Notably, Xist exhibited a preference for binding to active gene regions, as evidenced by the significantly higher expression of genes within the 10-kb range of Xist binding regions. The parallel upregulation of X-linked and autosomal genes in Δ RepB and male ES cells suggests a regulatory role of Xist in gene expression beyond its canonical function on the X chromosome. Therefore, the precise temporal modulation of Xist binding merits further investigation to elucidate its regulatory significance and during distinct stages of cell differentiation or development, particularly in understanding its autosomal targets which could have implications for gene regulation and cellular function.

Second, a recent study using an alternative Xist pulldown method, RAP-seq, also revealed the capability of Xist to bind to autosomes in naive human pluripotent stem cells (naive hPSCs) and mediate gene regulation²⁵. While we found that, on autosomes, Xist does not spread and instead covers only a narrow region corresponding to single genes, the previous study demonstrated broader regions of binding²⁵. Furthermore, our CHART experiments demonstrate

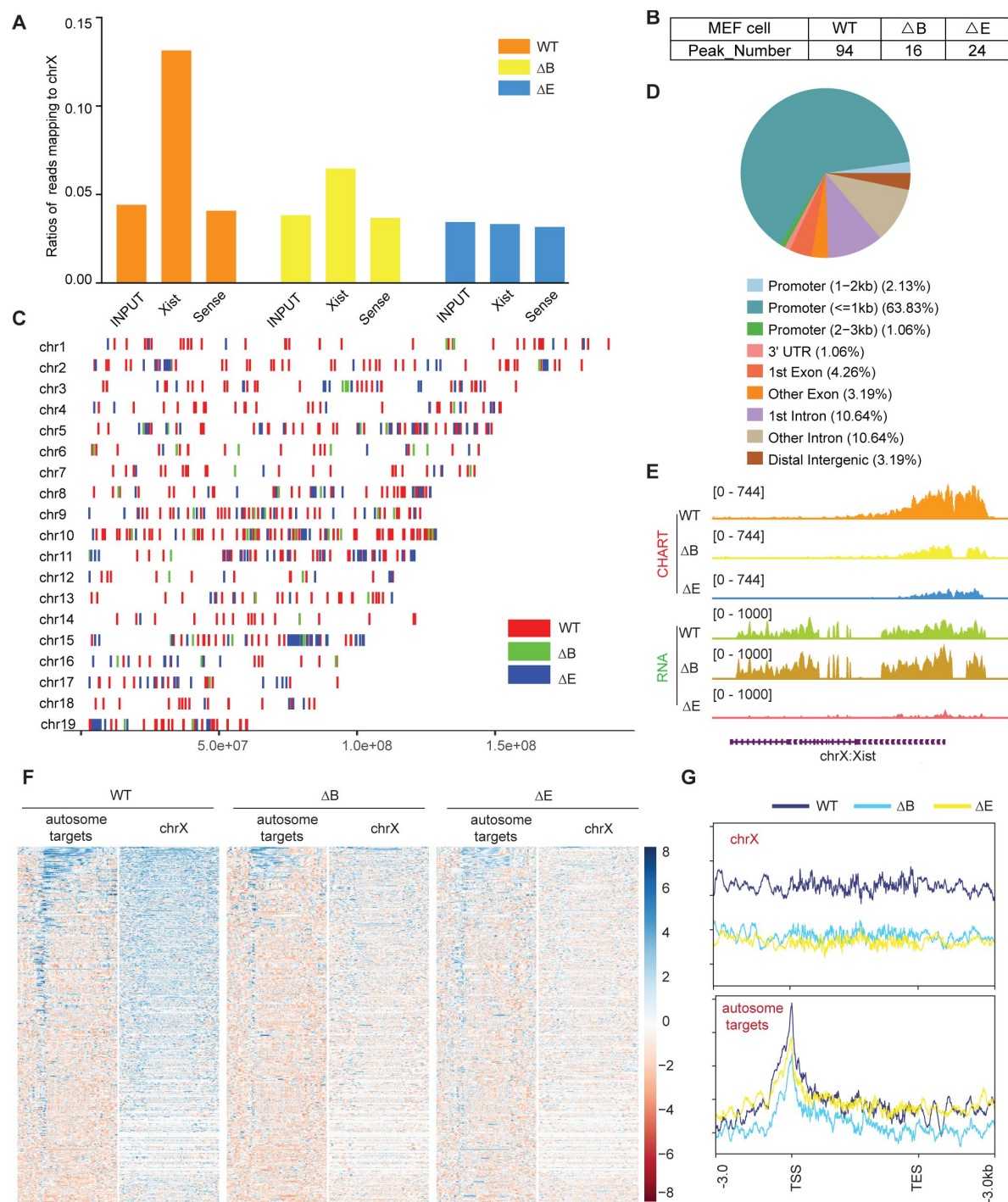


Figure 6.

Xist RNA also binds autosomal genes in post-XCI MEF cells.

- A.** Coverage of CHART-Seq reads (input, Xist, and sense control) on Chromosome X in WT, Δ RepB, and Δ RepE female MEF cells.
- B.** Xist peak number (MACS2 peak calling) on autosomes in WT, Δ RepB, and Δ RepE female MEF cells.
- C.** Xist peak patterns (MACS2 peak calling) on autosomes in WT, Δ RepB, and Δ RepE female MEF cells.
- D.** Feature annotation of Xist binding loci on autosomes in WT female ES cells.
- E.** Representative CHART-seq and RNA-seq track patterns of Xist in WT, Δ RepB, and Δ RepE female MEF cells.
- F-G.** Heatmap (F) and profile (G) depicting Xist coverage on autosome targets and chromosome X in WT, Δ RepB, and Δ RepE female MEF cells.

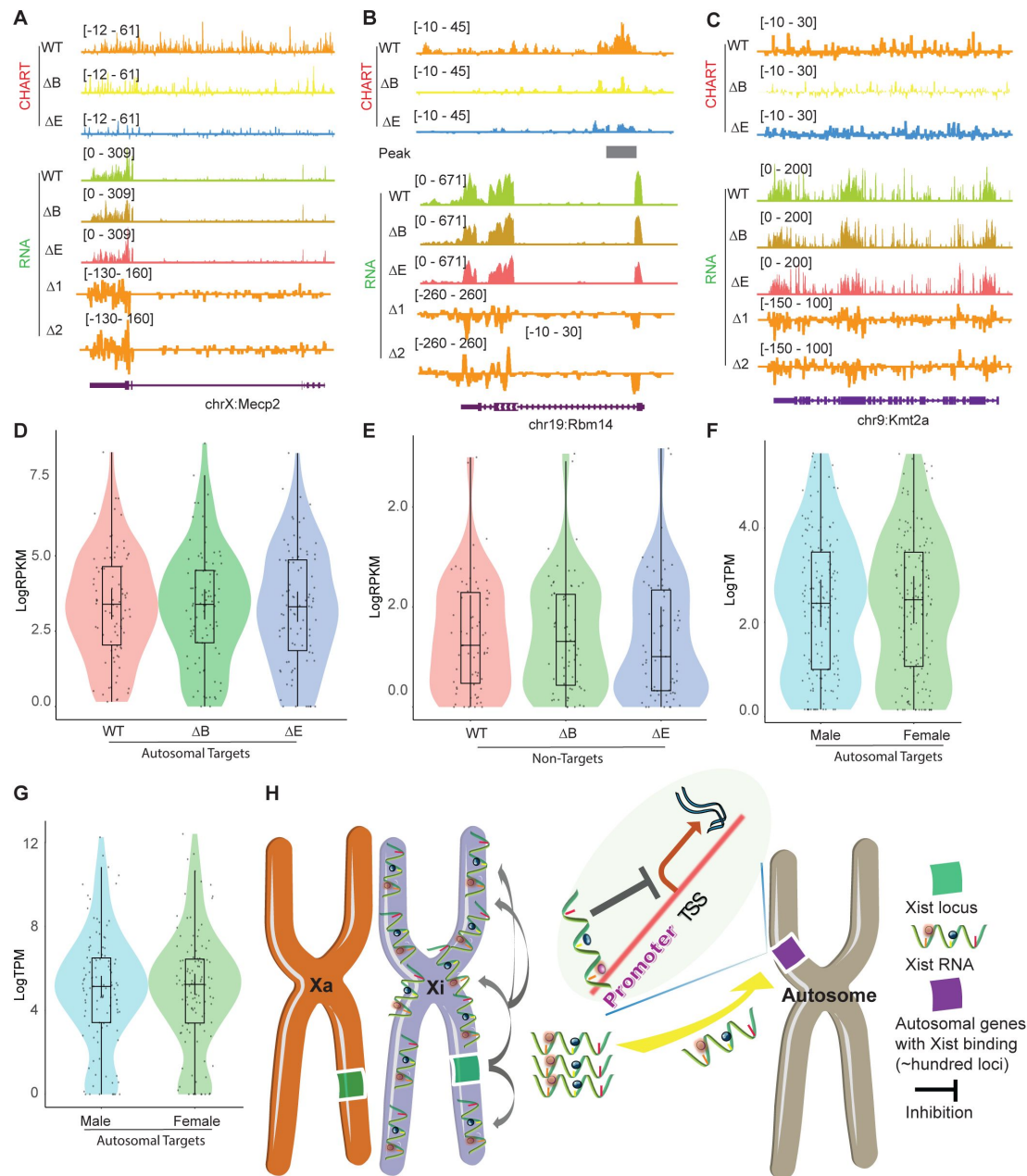


Figure 7.

Xist autosomal binding does not alter gene expression in MEF Cells.

A-C. Representative CHART-seq and RNA-seq track patterns of Xist autosomal binding genes such as *Mecp2* (A) and *Rbm14* (B), and autosome genes without Xist binding such as *kmt2a* (C) in WT, $\Delta RepB$, and $\Delta RepE$ female MEF cells. Change in coverage ($\Delta 1$ and $\Delta 2$) is shown below ($\Delta 1$ for $\Delta RepB$ - WT, and $\Delta 2$ for $\Delta RepE$ - WT ♀).

D-E. Gene expression levels for Xist targets (D) or non-targets (E) on autosomes in WT, $\Delta RepB$, and $\Delta RepE$ female MEF cells show no obvious changes.

F-G. Gene expression levels for Xist targets on autosomes in male and female MEF cells show no obvious changes.

H. Schematic of the Xist autosome binding pattern influences the gene expression. During the differentiation process of female ES cells, the Xist is specifically expressed from the Xist loci of one X chromosome (Xi). It then binds to the Xist loci of the Xi *in cis*, spreading across this chromosome and effectively silencing the expression of most of its genes. Additionally, some Xist complex will also bind to hundreds of loci on autosomes, thereby inhibiting the expression levels of target genes located there.

Xist binding to autosomes in post-XCI cells (e.g., MEF), whereas the previous study found no significant binding beyond the stem cell stage²⁵. Our analysis also indicates that Xist's autosomal targets tend to be Polycomb-depleted relative to neighboring genes, whereas the prior human study observed a positive correlation between XIST RNA and PRC2 marks. These discrepancies may arise from differences in the Xist pulldown methods (CHART versus RAP) employed by the two studies or from inherent differences between mouse and human systems.

Third, we considered the possibility that the binding of Xist to autosomes could merely be a consequence of Xist diffusion following saturation of binding sites on the Xi, rather than any programmed event during development. We are inclined to reject this notion and propose that Xist binding to autosomes is specifically programmed. Upon deleting RepB, the binding of Xist to the X chromosome weakens, but concomitantly, its binding to autosomal targets also diminishes (**Figure 2**). This suggests that Xist binding to autosomes is contingent upon Repeat B and is deliberate, rather than due to random Xist diffusion alone (in which case we would have expected increased autosomal binding). Additionally, following treatment with the Xist inhibitor X1, an increase in the expression of autosomal targets is observed (**Figure 5**), implying that the regulation of autosomes by Xist is not merely an overflow effect of Xist saturating the X chromosome.

In summary, our study advances understanding of Xist-mediated epigenetic regulation by highlighting unexplored interactions with autosomes. The identified correlations between Xist binding and gene expression involvement pose intriguing questions regarding the regulatory mechanisms governing these processes. These insights contribute to the evolving paradigm of Xist biology, underscoring the need for continued exploration of the complexity of epigenetic control mechanisms. Future investigations will delve deeper into the functional consequences of Xist binding on autosomes and explore the potential downstream effects on cellular differentiation, development, and diseases associated with its dysregulation, such as cancer, immunity, and neuron development^{25,44–48}. A comprehensive understanding of Xist's influence beyond X chromosomes is crucial, particularly laying the groundwork for future exploration of Xist as a potential therapeutic target.

Methods

Cell lines

The wild-type and Xist's Repeat B or E deletion MEF cell lines (M. musculus/M. castaneus F1 hybrid) has been previously described and generated³¹. Additionally, male Xist transgenic (Xist TG) MEF cells, previously denoted as "♂X+P"²³ were included in the study. MEFs were cultured in medium comprising DMEM, high glucose, GlutaMAX™ Supplement, pyruvate (Thermo Fisher Scientific), 10% FBS (Sigma), 25 mM HEPES pH 7.2-7.5 (Thermo Fisher Scientific), 1x MEM non-essential amino acids (Thermo Fisher Scientific), 1x Pen/Strep (Thermo Fisher Scientific), and 0.1 mM βME (Thermo Fisher Scientific), maintained at 37°C with 5% CO₂.

The wild-type and Xist's Repeat B deletion female ES cell line (M. musculus/M. castaneus F2 hybrid) harboring a mutated Tsix allele has been previously established^{31,49}. The male ES cell line was previously referred to as J1⁴⁹. ES cells were grown on γ-irradiated mouse embryonic fibroblast (MEF) feeder cells. Culture conditions included DMEM, high glucose, GlutaMAX™ Supplement, pyruvate (Thermo Fisher Scientific), 15% Hyclone FBS (Sigma), 25 mM HEPES pH 7.2-7.5, 1x MEM non-essential amino acids, 1x Pen/Strep, 0.1 mM βME, and 500 U/mL ESGRO recombinant mouse Leukemia Inhibitory Factor (LIF) protein (Sigma) at 37°C with 5% CO₂.

ES cell differentiation

Undifferentiated ES cells were initially cultivated on γ -irradiated MEF feeders for 3 days (day 0). Subsequently, the ES cell colonies were trypsinized (Thermo Fisher Scientific), and the feeders were removed. ES cells were then transitioned to a medium devoid of LIF and cultured in suspension for 4 days to form embryoid bodies (EBs). On day 4, the EBs were carefully settled onto gelatin-coated plates and allowed to undergo further differentiation until they were harvested on day 14.

CHART-seq

Xist CHART-seq methodology was conducted in accordance with a previously described protocol [50](#). Briefly, the cells were collected and suspended in PBS at a concentration of 2.5 million cells/mL. Subsequently, cross-linking was performed using 1% formaldehyde at room temperature for 10 min, followed by quenching with 0.125 M glycine for an additional 10 min. After two washes with ice-cold PBS, cells were pelleted and snap-frozen in liquid nitrogen. For CHART, 25 million cells were thawed on ice and resuspended in 1 mL ice-cold sucrose buffer (10 mM HEPES pH 7.5, 0.3 M sucrose, 1% Triton X-100, 100 mM potassium acetate, 0.1 mM EGTA, 0.5 mM spermidine, 0.15 mM spermine, 1 mM DTT, 1x protease inhibitor cocktail, 10 U/mL SUPERase•In™ RNase Inhibitor). The cell suspension was rotated at 4°C for 10 min, followed by dilution with 2 mL of cold sucrose buffer. Douncing was performed using an RNaseZAP treated 15-mL glass Wheaton Dounce tissue grinder 20 times with a tight pestle. The nuclear suspension was carefully layer on top of a cushion of 7.5 ml glycerol buffer (10 mM HEPES pH 7.5, 25% glycerol, 1 mM EDTA, 0.1 mM EGTA, 100 mM potassium acetate, 0.5 mM spermidine, 0.15 mM spermine, 1 mM DTT, 1x cComplete EDTA-free protease inhibitor cocktail, and 5 U/mL RNase inhibitor) in a new 15 ml tube, and centrifuged at 1,500 g for 10 min at 4°C. The pellet was resuspended in 3 mL of PBS and cross-linked with 3 ml 6% formaldehyde for 30 min at room temperature with rotation. Afterwards, nuclei were pelleted by centrifugation at 1,000 g for 5 min at 4°C and washed three times with ice-cold PBS, and resuspended in 1 mL ice-cold nuclear extraction buffer (50 mM HEPES, pH 7.5, 250 mM NaCl, 0.1 mM EGTA, 0.5% N-lauroylsarcosine, 0.1% sodium deoxycholate, 5 mM DTT, 10 U/mL RNase inhibitor), rotated for 10 min at 4°C, and then centrifuged at 400 g for 5 min at 4°C and resuspended in 230 μ L cold sonication buffer (50 mM HEPES pH 7.5, 75 mM NaCl, 0.1 mM EGTA, 0.5% N-lauroylsarcosine, 0.1% sodium deoxycholate, 0.1% SDS, 5 mM DTT, 10 U/mL RNase inhibitor) to a final volume of $\sim$ 270 μ L. The nuclei were sonicated in a microtube using a Covaris E220 sonicator (140 W peak incident power, 10% duty factor, 200 cycles/burst, 4°C, 300 s). Sonicated chromatin was centrifuged at 16,000 g for 20 min at 4°C, and transfer the supernatant ($\sim$ 220 μ L) to new tube and add 170 μ L sonication buffer to a final volume of $\sim$ 390 μ L, which was pre-cleared by 60 μ L MyOne Streptavidin C1 beads (Thermo Fisher Scientific) in 640 μ L 2x hybridization buffer (50 mM Tris pH 7.0, 750 mM NaCl, 1% SDS, 1 mM EDTA, 15% formamide, 1 mM DTT, 1 mM PMSF, 1x cComplete EDTA-free protease inhibitor cocktail, 100 U/mL RNase inhibitor) at room temperature for 1 h with rotation. Pre-cleared chromatin was divided into two CHART reactions for Xist and control capture, and 1% was saved as an input sample. For each CHART reaction, 36 pmol of antisense (Xist) or sense (control) biotinylated capture probes (pooled probe as previously described) were used [21](#). Hybridization was performed at room temperature with rotation overnight, and 120 μ L C1 beads were added to the samples and incubated at 37°C for 1 h with rotation. The beads were washed once with 1x hybridization buffer (33% sonication buffer, 67% 2x hybridization buffer) at 37°C for 10 min, five times with wash buffer (10 mM HEPES pH 7.5, 150 mM NaCl, 2% SDS, 2 mM EDTA, 2 mM EGTA, 1 mM DTT) at 37°C for 5 min, and twice with elution buffer (10 mM HEPES pH 7.5, 150 mM NaCl, 0.5% NP-40, 3 mM MgCl₂, 10 mM DTT) at 37°C for 5 min. 1% of the final wash was saved as an “RNA pulldown” sample. CHART-enriched DNA was eluted twice in 200 μ L of elution buffer supplemented with 5 U/ μ L RNase H (New England BioLabs) at room temperature for 20 min. The “input” sample and CHART DNA were treated with 0.5 mg/mL RNase A (Thermo Fisher Scientific) at 37°C for 1 hr with rotation and then incubated with 1% SDS, 10 mM EDTA, and 0.5 mg/mL proteinase K (Sigma) at 55°C for 1 hr. Reverse

crosslinking was performed using 150 mM NaCl (final concentration 300 mM) at 65°C overnight. DNA was purified using phenol-chloroform and further sheared to ~300bp fragments using Covaris E220e (140 W peak incident power, 10% duty factor, 200 cycles/burst, 120 s, 4°C). Sonicated DNA was purified using 1.8x Agencourt AMPure XP beads (Beckman Coulter). Input and CHART DNA libraries were prepared according to the protocol of the NEBNext® Ultra™ II DNA Library Prep Kit for Illumina (NEB E7645S). Libraries were sequenced on Novaseq S4, generating approximately 30 million 150-nt paired-end reads per sample.

Data source and processing

Raw data for CHART-seq and bulk RNA-seq generated in this study have been deposited in the Gene Expression Omnibus (GEO) database with accession number GSE271096 and GSE271097, respectively. ChIP-seq data utilized in this study were sourced from a study conducted by David et al. in 2020 ³² with GEO accession number: GSE135389. Additionally, RNA-seq and ChIP-seq data specifically pertaining to Xist inhibitor X1 were obtained from a dataset published in 2022 ³⁵ with GEO accession number: GSE141683. RNA-seq data for Xist TG in mES cells were acquired from another dataset published in 2017 ³⁴ with GEO accession number: GSE92894. RNA-seq data for male and female MEF cells were acquired from two published dataset with GEO accession number: GSE246699 and GSE118443. The data were processed and re-analyzed consistently in this study.

CHART-seq and ChIP-seq analysis

Initial data preprocessing involved removing adaptors using trim_galore/cutadapt (versions 0.4.3/1.7.1). Subsequently, trimmed reads were prepared for alignment. Alignment of the genomes of *Mus musculus* (mus) and *Mus castaneus* (cas) was performed using NovoAlign (version 4.03). Aligned reads were then mapped back to the reference mm10 genome using SNPs to generate a BAM file ²⁷. Samtools (version 1.11) was employed for random sampling to ensure uniform library sizes across the samples. Utilizing DeepTools (version 3.1.2), input-subtracted ChIP and CHART coverage profiles were created, resulting in bigwig. To discern the peaks of Xist CHART on the autosomes, chrX reads were initially excluded from the BAM file. MACS2 (version 2.1) was used for peak calling using default parameters. The resultant peak files from two replicates were merged using bedtools (version 2.30) intersect. Bedtools window (version 2.30) facilitated the creation of gene lists based on peak sizes, spanning 10, 20, 50, and 100 kilobases. Deeptools was used to conduct comparative analyses of coverage, represented through either profile or heatmap plots, for genomic regions associated with Xist, H3K27me3, or H2A119ub across distinct categories of gene lists.

RNA-seq

Total RNA was extracted using TRIzol (Thermo Fisher Scientific), and rRNA was selectively removed using the NEBNext rRNA Depletion Kit (New England BioLabs), according to the manufacturer's instructions. RNA-seq libraries were prepared using the NEBNext Ultra II Directional RNA Library Prep Kit for Illumina (New England BioLabs), which enabled the creation of strand-specific RNA-seq libraries. Libraries were sequenced on Novaseq S4, generating ~30 million 150-nt paired-end reads per sample.

RNA-seq analysis

Raw RNA-seq data underwent adaptor removal and trimming using trim_galore/cutadapt (versions 0.4.3/1.7.1). RNA-seq data were mapped to three different genomes: C57BL/6J (mm10), *M. musculus* (mus), and *M. castaneus* (cas). By employing Subread (version 2.0.2) featureCounts ⁵¹, counts per gene were computed, while deeptools were utilized to generate bigwig files, providing a comprehensive overview of read coverage across the genomes. For expression quantification, both Reads Per Kilobase Million (RPKM) and log transformed RPKM values were calculated to establish a robust foundation for downstream analysis.

Quantification and Statistical Analysis

In the experimental design, two replicates were used for both the CHART and RNA-seq analyses. Statistical analyses were conducted, and the corresponding p-values are transparently reported in the figures and their respective legends. The significance levels are denoted by asterisks, where *, **, and *** represent $p < 0.05$, $p < 0.01$, and $p < 0.001$, respectively.

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Statement of financial interest

JTL is an advisor to Skyhawk Therapeutics, a cofounder of Fulcrum Therapeutics, and a non-executive Director of the GSK.

Supplementary figures

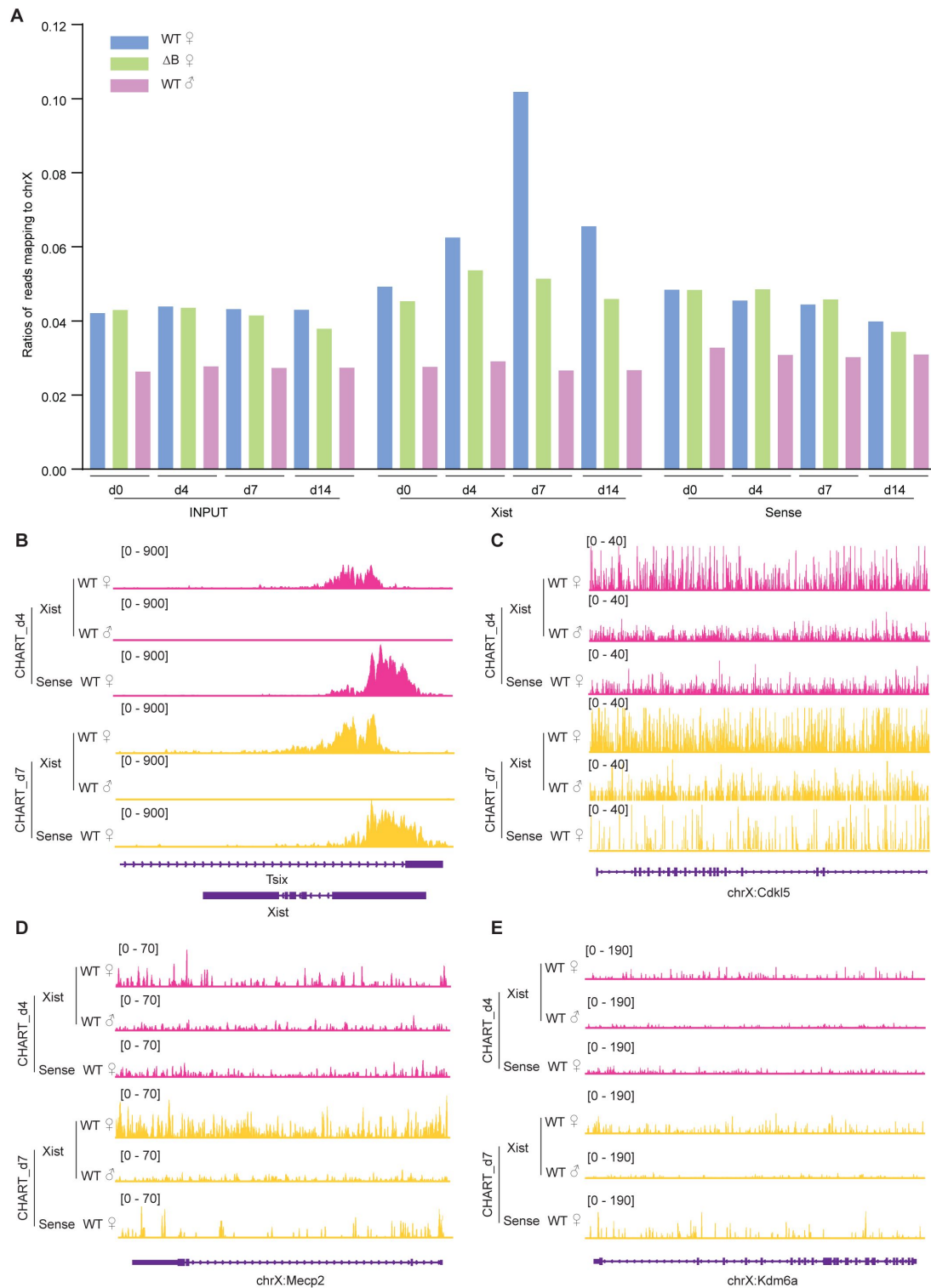


Figure S1.

Xist binds select autosomal genes in trans.

A. Xist CHART-seq reads percent of autosomes and Chromosome X in WT and ΔRepB female ES cells. ES day 7 exhibits the highest Xist coverage. Sense probe and male ES cells are used as a control. **B-E.** Representative binding signals of Xist RNA on Xist locus (B), genes such to XCI such as *Cdkl5* (C), *Mecp2* (D), and escapee genes (*Kdm6a*) (E) in WT female ES cells on day 4 and day 7. WT male ES cells and sense probe are used as control.

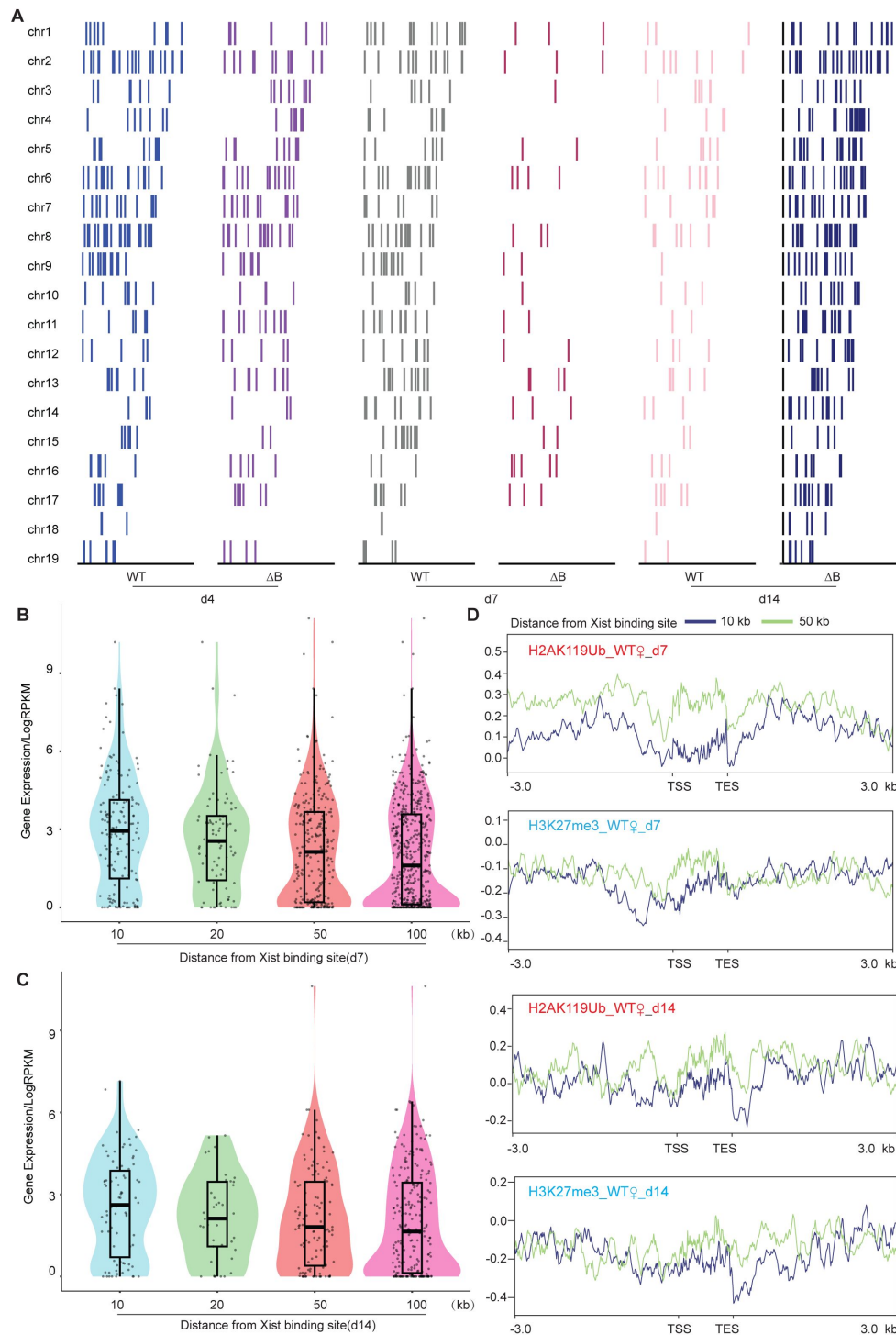


Figure S2.

Genes bound by Xist exhibit higher expression levels and lower H3K27me3 and H2AK119ub binding levels.

A. Xist peak pattern (MACS2 peak calling) on autosomes in WT and $\Delta RepB$ female ES cells.

B-C. The analysis of gene expression within the 10, 20, 50, and 100-kilobase binding regions of Xist is performed in WT female ES cells at day 7 (B), and day 14 (C), respectively.

D. Profile plots showing H3K27me3 and H2AK119ub coverage over genes within the 10 and 50-kilobase binding regions of Xist in WT female ES cells.

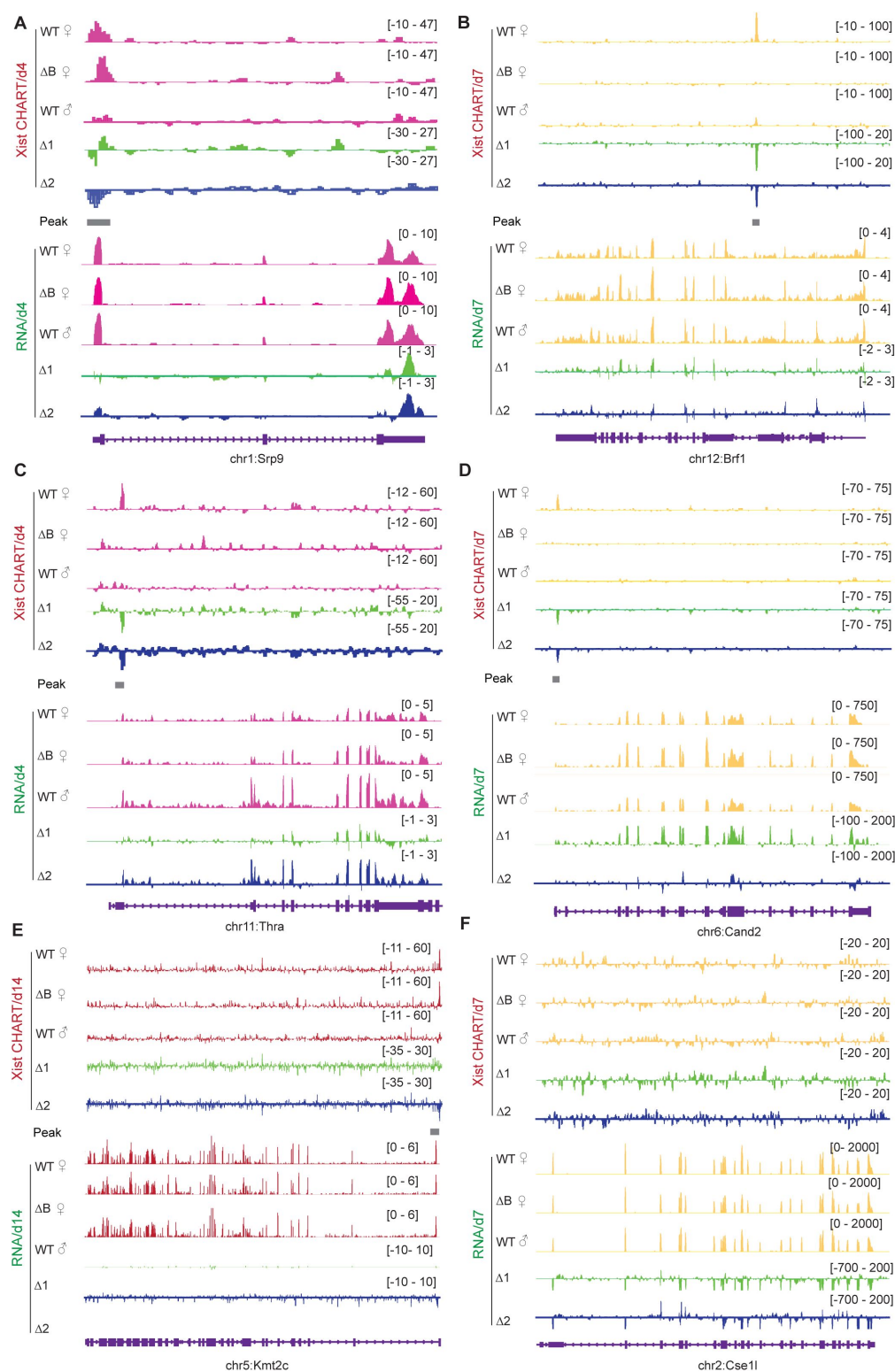


Figure S3.

Example of Xist binding on autosomal genes and influence on gene expression.

This figure illustrates CHART-seq and RNA-seq patterns of autosomal genes, including *Srp9*, *Brf1*, *Thra*, *Cand2*, and *Kmt2c*, which exhibit Xist binding on different days. *Ces1l*, which lacks Xist binding, is used as a control. Change in coverage ($\Delta 1$ and $\Delta 2$) is shown below ($\Delta 1$ for $\Delta \text{RepB}^- - \text{WT}^-$, and $\Delta 2$ for $\text{WT}^\sigma - \text{WT}^-$).

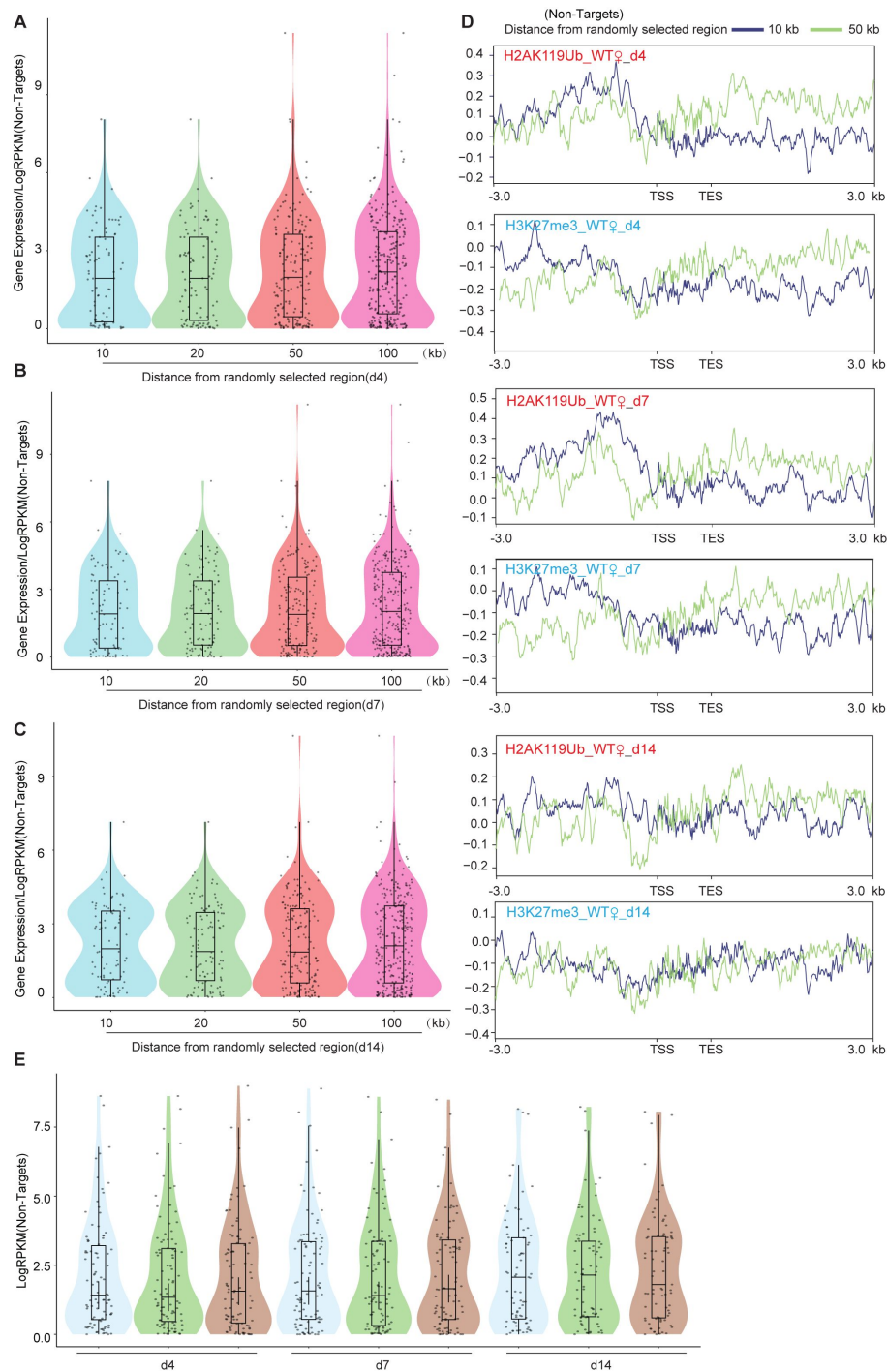


Figure S4.

Genes not bound by Xist exhibit no changes in gene expression or differences in H3K27me3 and H2AK119ub signals.

A-C. The analysis of gene expression within the 10, 20, 50, and 100-kilobase randomly selected regions is performed in WT female ES cells at day 4 (A), day 7 (B), and day 14 (C), respectively.

D. Profile plots showing H3K27me3 and H2AK119ub coverage over genes within the 10 and 50-kilobase randomly selected regions in WT female ES cells at different time points.

E. Assessing gene expression levels of Xist non-targets on autosomes in WT, Δ RepB female ES cells, and male ES cells at different time points.

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Reviewer #1 (Public review):

Summary:

The manuscript by Yao S. and colleagues aims to monitor the potential autosomal regulatory role of the master regulator of X chromosome inactivation, the Xist long non-coding RNA. It has recently become apparent that in the human system, Xist RNA can not only spread in cis on the future inactive X chromosome but also reach some autosomal regions where it recruits transcriptional repression and Polycomb marking. Previous work has also reported that Xist RNA can show a diffused signal in some biological contexts in FISH experiments.

In this study, the authors investigate whether Xist represses autosomal loci in differentiating female mouse embryonic stem cells (ESCs) and somatic mouse embryonic fibroblasts (MEFs). They perform a time course of ESC differentiation followed by Capture Hybridization of Associated RNA Targets (CHART) on both female and male ESCs, as well as pulldowns with sense oligos for Xist. The authors also examine transcriptional activity through RNA-seq and integrate this data with prior ChIP-seq experiments. Additional experiments were conducted in MEFs and Xist-ΔB repeat mutants, the latter fails to recruit Polycomb repressors.

Based on this experimental design, the authors make several bold claims:

- (1) Xist binds to about a hundred specific autosomal regions.
- (2) This binding is specific to promoter regions rather than broad spreading.
- (3) Xist autosomal signal is inversely correlated with PRC1/2 marks but positively correlated with transcription.
- (4) Xist targeting results in the attenuation of transcription at autosomal regions.
- (5) The B-repeat region is important for autosomal Xist binding and gene repression.
- (6) Xist binding to autosomal regions also occurs in somatic cells but does not lead to gene repression.

Together, these claims suggest that Xist might play a role in modulating the expression of autosomal genes in specific developmental and cellular contexts in mice.

Strengths:

This paper deals with an interesting hypothesis that Xist ncRNA can also function at autosomal loci.

Weaknesses:

The claims reported in this paper are largely unsubstantiated by the data, with multiple misinterpretations, lacking controls, and inadequate statistics. Fundamental flaws in the

experimental design/analysis preclude the validity of the findings. Major concerns are listed below:

(1) The entire paper is based on the CHART observation that Xist is specifically targeted to autosomal promoters. Overall, the data analysis is flawed and does not support such conclusions. Importantly the sense WT and the 0h controls are not used, nor are the biological replicates. Data is typically visualized without quantification, and when quantified, control loci/gene sets are erroneously selected. Firstly, CHART validation on the X in FigS1 is misleading and not based on any quantifications (e.g., see the scale on Kdm6a (0-190) compared to Cdkl5 (0-40)). If scaled appropriately, there is Xist signal on the escapee. All X-linked loci should have been quantified and classified based on escape status; sense control should also be quantified, and biological replicates should be shown separately. Secondly, and most importantly, Figure 1 does not convincingly show specific Xist autosomal binding. Panel A quantification is on extremely variable y-scales and actually shows that Xist is recruited globally to nearly all autosomal genes, likely indicating an unspecific signal. Again, the sense and 0h controls should have been quantified along with biological replicates. Upon inspecting genome browser tracks of all regions reported in the manuscript (Rbm14, Srp9, Brf1, Cand2, Thra, Kmt2c, Kmt2e, Stau2, and Bcl7b), the signal is unspecific on all sites with the possible exception of Kmt2e. On all other loci, there is either a strong signal in the 0h ESC controls or more signal in some of the sense controls. This implies that peak calling is picking up false positive regions. How many peaks would have been picked up if the sense or the 0h controls were used for peak calling? It is likely that there would be a lot since there are also possible "peaks" (e.g., Fzd9) in control tracks. Further inspection of the data was not possible as the authors did not provide access to the raw fastq files. When inspecting results from past published experiments {Engreitz, 2013 #1839} reported regions were not bound by Xist. Thirdly, contrary to the authors' claim, deleting the B repeat does not lead to a loss of autosomal signal. Indeed, comparing Fig1A and Fig2B side by side clearly shows no difference in the autosomal signal, likely because the autosomal signal is CHART background. Properly quantifying the signal with separate replicates as well as the sense and 0h controls is vital. Overall current data together with published results indicate that CHART peak calling on autosomes is due to technical noise or artefacts.

(2) The RNA-seq analysis is also flawed and precludes strong statements. Firstly, the analysis frequently lacks statistical analysis (Fig3B, FigS2B-C) and is often based on visualizations (Fig 3D-G) without quantifications. Day 4 B-repeat deletion does not lead to a significant change in the expression of genes close to Xist signal (Fig3H, d14 does not fully show). Secondly, for all transcriptional analysis, it is important to show autosomal non-target genes, which is not always done. Indeed, both males and B repeat deletion will lead to transcriptional changes on autosomes as a secondary effect from different X inactivation status. The control set, if used, is inappropriate as it compares one randomly selected set of ~100 genes. This introduces sampling error and compares different classes of genes. Since Xist signal targets more active genes, it is important to always compare autosomal target genes to all other autosomal genes with similar basal expression patterns.

(3) The ChIP-seq analysis also has some problems. The authors claim that there is no positive correlation between genes close to Xist autosomal binding (10kb) compared to those 50kb away (Fig 3C, S2D); however, this analysis is based entirely on metagene visualization. Signal within the Xist binding sites should be quantified (not genes close by) and compared to other types of genomic loci and promoters. Focusing on the 50kb group only as controls is misleading. Secondly, the authors only look at PRC mark signal upon differentiation; what about the 0h timepoint, i.e., is there pre-marking? Most worryingly, the data analysis is not consistent between figures (see Fig3C vs 5H-I). In Fig5, the group of Xist targets was chosen as those within 100kb of Xist binding, which would encompass all the control regions from Fig3C. In this analysis, the authors report that there is Xist-dependent H3K27me3 deposition,

and in fact, here the Xist autosomal targets have more of it than the controls. Overall, all of this analysis is misleading, and clear conclusions cannot be made.

All in all, because the fundamental observation is not robust (see point 1), all subsequent analyses are also affected. There are also multiple other inconsistencies within the analysis; however, they have not been included here for brevity.

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Reviewer #2 (Public review):

Summary:

To follow-up on recent reports of Xist-autosome interaction the authors examine female (and male transgenic) mESCs and MEFs by CHARTseq. Upon finding that only 10% of reads map to X, they sought to identify reproducible alternative sites of Xist-binding, and identify ~100 autosomal Xist-binding sites and show a transient impact on expression.

Strengths:

The authors address a topical and interesting question with a series of models including developmental timepoints and utilize unbiased approaches (CHARTseq, RNAseq). For the CHARTseq they have controls of both sense probes and male cells; and indeed do detect considerable background with their controls. The use of deletions emphasizes that intact functional Xist is involved. The use of 'metagene' plots provides a visual summation of genic impact.

Weaknesses:

Overall, the result presentation has many 'sample' gene presentations (in contrast to the stronger 'metagene' summation of all genes). The manuscript often relies on discussion of prior X chromosomal studies, while the data generated would allow assessment of the X within this study to confirm concordance with prior results using the current methodology/cell lines. Many of the 'follow-up' analyses are in fact reprocessing and comparison of published datasets. The figure legends are limited, and sample size and/or source of control is not always clear. While similar numbers of autosomal Xist-binding sites were often observed, the presented data did not clarify how many were consistent across time-points/cell types. While there were multiple time points/lines assessed, only 2 replicates were generally done.

Aim achievement:

The authors do identify autosomal sites with enrichment of chromatin marks and evidence of silencing. More details regarding sample size and controls (both treatment, and most importantly choice of 'non-targets' - discussed in comments to authors) are required to determine if the results support the conclusions.

Specific scenarios for which I am concerned about the strength of evidence underlying the conclusion:

I found the conclusion "Thus, RepB is required not only for Xist to localize to the X-chromosome but also for its localization to the ~100 autosomal genes " (p5) in contrast to the statement 2 lines prior: "A similar number of Xist peaks across autosomes in Δ RepB cells was observed and the autosomal targets remained similar". Some quantitative statistics would assist in determining impact, both on autosomes and also X; perhaps similar to the quintile analysis done for expression.

It is stated that there is a significant suppression of X-linked genes with the autosomal transgenes; however, only an example is shown in Figure 4B. To support this statement, a full X chromosomal geneset should be shown in panels F and G, which should also list the number of replicates. As these are hybrid cells, perhaps allelic suppression could be monitored? Is Med14 usually subject to X inactivation in the Ctrl cells, and is the expression reduced from both X chromosomes or preferentially the active (or inactive) X chromosome?

The expression change for autosomes after transgene induction is barely significant; and it was not clear what was used as the Ctrl? This is a critical comparator as doxycycline alone can change expression patterns.

In the discussion there is the statement. "Genetic analysis coupled to transcriptomic analysis showed that Xist down-regulates the target autosomal genes without silencing them. This effect leads to clear sex difference - where female cells express the ~100 or so autosomal genes at a lower level than male cells (Figure 7H)." This sweeping statement fails to include that in MEFs there is no significant expression difference, in transgenics only borderline significance, and at d14 no significant expression difference. The down-regulation overall seems to be transient during development while targeting is ongoing?

Finally, I would have liked to see discussion of the consistency of the identified genes to support the conclusion that the autosomal sites are not merely the results of Xist diffusion.

The impact of Xist on autosomes is important for consideration of impact of changes in Xist expression with disease (notably cancers). Knowing the targets (if consistent) would enable assessment of such impact.

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Reviewer #3 (Public review):

Summary:

Yao et al use CHART to identify chromatin associated with Xist in female mouse ESCs, and, as control, male ESCs at various timepoints of differentiation. Besides binding of Xist to X chromosome regions they found significant binding to autosomes, concentrating mostly on promoter regions of around 100 autosomal genes, as elucidated by MACS. The authors went on to show that the RepB repeat is mostly responsible for these autosomal interactions using a female ESC line in which RepB is deleted. Evidence is provided that Xist interacts with active autosomal genes containing lower coverage of repressive marks H3K27me3 and H2AK119ub and that RepB dependent Xist binding leads to dampening of expression, but not silencing of autosomal genes. These results were confirmed by overexpression studies using transgenic ESCs with doxycycline-inducible Xist as well as via a small molecule inhibitor of Xist (X1), inducing/inhibiting the dampening of autosomal genes, respectively. Finally, using MEFs and Xist mutants RepB or RepE the authors provide evidence that Xist is bound to autosomal genes in cells after the XCI process but appears not to affect gene expression. The data presented appear generally clear and consistent and indicate some differences between human and mouse autosomal regulation by Xist.

Strengths:

Regulation of autosomal gene expression by Xist is a "big deal" as misregulation of this lncRNA causes developmental defects and human disease. Moreover, this finding may explain sex-specific developmental differences between the sexes. The results in this manuscript identify specific mouse autosomal genes bound by Xist and decipher critical Xist regions that mediate this binding and gene dampening. The methods used in this study are

appropriate, and the overall data presented appear convincing and are consistent, indicating some differences between human and mouse autosomal regulation by Xist.

Weaknesses:

(1) The figure legends and/or descriptions of data are often very short lacking detail, and this unnecessarily impedes the reading of the manuscript, in particular the figures would benefit not only from more detailed descriptions/explanations of what has been done but also what is shown. This will facilitate the reading and overall comprehension by the reader. One out of many examples: In Fig S1B in the CHART data at d4 and d7 there is not only signal in female WT Xist antisense but also in female sense control. For a reader that is not an expert in XCI it would be helpful to point out in the legend that this signal corresponds to the lncRNA Tsix (I suppose), that is transcribed on the other strand.

(2) Different scales are used in the lower panels of Figures 1A and 2A, which makes it difficult to directly compare signals between the different differentiation stages.

(3) In this study some of the findings on mouse cells contrast previously published results in human ESCs: 1) Xist binding occurs preferentially to promoters in mice, not in human. 2) Binding of Xist is mostly detected in polycomb-depleted regions in mice but there is a positive correlation between Xist RNA and PRC2 marks in human ESCs. These differences are surprising but may be very interesting and relevant. While I am aware that this might be a difficult task, it would be helpful to experimentally address this issue in order to distinguish whether species specific and/or methodological differences between the studies are responsible for these differences.

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